# Convolutional Neural Networks for Multimodal Remote Sensing Data Classification

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***Abstract*— In recent years, enormous research has been made to improve the classification performance of single-modal remote sensing (RS) data. However, with the ever-growing availability of RS data acquired from satellite or airborne platforms, simultaneous processing and analysis of multimodal RS data pose a new challenge to researchers in the RS community. To this end, we propose a deep-learning-based new framework for multimodal RS data classification, where convolutional neural networks (CNNs) are taken as a backbone with an advanced cross-channel reconstruction module, called CCR-Net. As the name suggests, CCR-Net learns more compact fusion representations of different RS data sources by the means of the reconstruction strategy across modalities that can mutually exchange information in a more effective way. Extensive experiments conducted on two multimodal RS datasets, including hyperspectral (HS) and light detection and ranging (LiDAR) data, i.e., the Houston2013 dataset, and HS and synthetic aperture radar (SAR) data, i.e., the Berlin dataset, demonstrate the effectiveness and superiority of the proposed CCR-Net in comparison with several state-of-the-art multimodal RS data classification methods. The codes will be openly and freely available at https://github.com/danfenghong/IEEE_TGRS_CCR-Net for the sake of reproducibility.**

***Index Terms*— Classification, convolutional neural networks (CNNs), cross-channel, hyperspectral (HS), light detection and ranging (LiDAR), multimodal, reconstruction, remote sensing (RS), synthetic aperture radar (SAR).**

## I. Introduction

With the rapid development of aeronautics and astronautics, remote sensing (RS) techniques have been playing a vital role in a variety of Earth observation (EO) tasks, such as land cover classification [1]–[3], scene classification [4], [5], mineral exploration and mapping [6], [7], object/target detection [8]–[10], environmental monitoring [11], urban planning [12]–[14], biodiversity conservation [15], and disaster response and management [16].

Over the past decade, research on land cover mapping and inversion mainly focuses on the algorithm development of RS data (acquired by the single sensor) processing and analysis [17]. For example, Kang *et al.* [18] first proposed to use convolutional sparse coding (CSC) on complex domains to restore the interferometric phase of synthetic aperture radar (SAR) data. Wu *et al.* [19] developed spatial-frequency channel features with the application to geospatial object detection using multispectral (MS) RS data. Huang *et al.* [20] utilized the local manifold learning to extract geometric representations from light detection and ranging (LiDAR) for point cloud classification. Hong *et al.* [21] designed a novel invariant attribute profile (IAP) for hyperspectral (HS) image classification, achieving state-of-the-art results on several popular datasets and showing the effectiveness of the proposed method.

However, only using single EO data, e.g., HS data, SAR data, LiDAR data, and MS data, inevitably meets the performance bottleneck in identifying and recognizing objects of interest due to the difficulties in excavating and jointly using the information potential of multimodal heterogeneous data. For this reason, the ever-growing availability of RS data can provide the possibility to acquire more diverse information to further help identify or classify the materials more accurately. Fig. 1 shows a showcase to clarify the potential and advantage of using multimodal RS data (e.g., HS and LiDAR data) over only using single modalities in the land cover classification task. Simultaneously, this also poses a great challenge to developing more advanced multimodal data processing and analysis approaches. There have been some traditional multimodal algorithms proposed in recent years, providing a good starting point to the challenging field.

Morphological profiles (MPs) and subspace learning-based approaches are two main groups for feature extraction and classification of multimodal RS data. In [22], the MPs of HS and LiDAR data are fused on a non-Euclidean space, i.e., manifolds, via graph-based subspace embedding methods. Furthermore, Ghamisi *et al.* [23] proposed to jointly extract the attribute profiles (APs) instead of MPs from HS and LiDAR for land cover classification. Xia *et al.* [24] developed an ensemble classifier with the joint use of HS and LiDAR data for the classification task. In the subspace learning-based group, Camps-Valls *et al.* [25] presented a general kernel framework for multitemporal and multisource RS data

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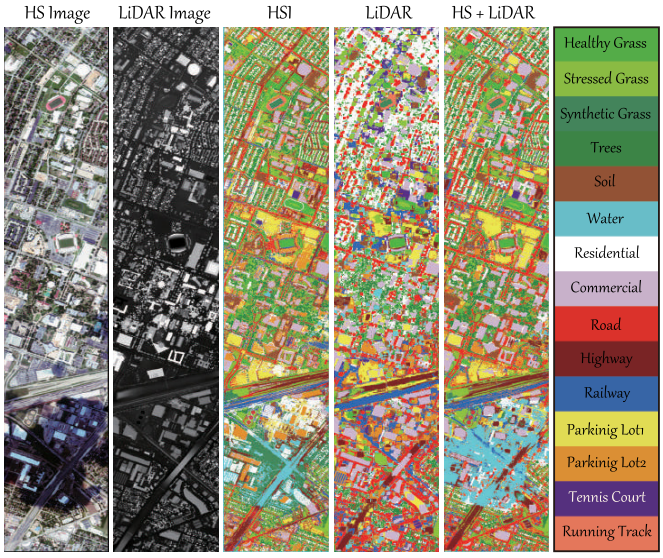



Fig. 1. Showcase to clarify the potential and advantage of using multimodal RS data (e.g., HS image + LiDAR) over only using single modalities (e.g., HS image or LiDAR) in the classification task.

classification and change detection. Yan *et al.* [26] measured the similarities of multisource RS data in both Euclidean and angular distance-based embeddings. Hong *et al.* [27] jointly performed the common subspace learning and regression on HS–MS correspondences. Such a shared subspace is capable of better transferring and interacting information of different modalities, which is applicable to both multimodal and crossmodal RS data classification tasks. They also discussed different regression regularization terms, e.g., $\ell_2$-norm and $\ell_1$-norm, and showed the superiority of $\ell_1$-norm regression in multimodal feature extraction [28]. The same investigators further extended their work to a semisupervised version by learning to align manifold structure between two modalities [29] for land cover and land use classification. Moreover, Hu *et al.* [30] deeply investigated the topological analysis theory and developed a novel MAPPER-based semisupervised fusion algorithm, which is extended by manifold alignment framework [31], for HS and polarimetric SAR images. Very recently, a novel shared and specific feature learning model is developed in [32] for multimodal land cover classification, where several multimodal RS benchmark datasets are provided and will be freely and openly available for the community. Despite the desirable performance obtained by these traditional shallow models, the feature representation ability still remains limited due to a relatively large gap between different modalities.

Inspired by the recent success of various deep networks in extracting more discriminative features from the data, e.g., convolutional neural networks (CNNs), recurrent neural networks (RNNs) [33], graph convolutional networks (GCNs) [34], and deep similarity network [35], some preliminary multimodal networks have been proposed and used for RS images to break the performance bottleneck and obtain better classification results than that only using single modalities [36]. It is worth noting that the fusion strategy is a key factor in determining the performance of multimodal networks, which can be roughly divided into two groups: concatenation-based fusion and alignment-based fusion. In general, the former one directly stacks images or features in early, middle, or late stages [37], leading to early fusion, middle fusion, and late fusion, respectively, while the latter one, as the name suggests, aligns different modalities by similarity measurements or constraints and achieves the fusion process effectively. More specifically, Hang *et al.* [38] integrated the HS and LiDAR data in an advanced cascade fashion by learning a coupled CNN, named Co-CNN. Motivated by the multimodal deep learning work in [39], Hong *et al.* [40] developed a simple and effective encoder–decoder network, called EndNet, for classification of HS and LiDAR data. Hong *et al.* [41] further modified this framework by introducing mutual and adversarial fusion strategies, yielding a semisupervised cross-modal deep network. Gadiraju *et al.* [42] developed a multimodal deep learning framework for crop classification using the MS and multitemporal satellite images. Furthermore, Suel *et al.* [43] also utilized the multimodal deep learning techniques to measure income, overcrowding, and environmental deprivation in urban areas by the means of RS data and street-view images. These newly developed approaches have been proven to be effective in fusing multiple RS data sources, yet the ability of these existing methods in handling multimodal data, particularly heterogeneous data (e.g., HS and SAR data, HS, and LiDAR), remains limited. This could be a lack of more advanced fusion strategies to better eliminate the gap between different modalities and obtain diagnostic features.

To overcome this difficulty, a novel cross-channel reconstruction (CCR) mechanism is designed and embedded into a CNN-based (CNN) framework, CCR-Net for short, for multimodal RS data classification. The CCR is a plug-and-play module, which can be easily embedded in various backbone networks, e.g., our CNNs. Moreover, the CCR module is capable of learning more compact feature representations, even though from heterogeneous data, by reconstructing different modalities or their features in a cross-channel fashion. In detail, our contributions to this article can be summarized as follows.

1) We develop a novel CNN-dominated multimodal RS data classification backbone framework with a well-designed CCR module, called CCR-Net.
2) We design a plug-and-play CCR module, aiming to blend the multimodal features in a more compact way. As the name suggests, the CCR module reconstructs the multimodal representations and outputs its cross-channel form to achieve effective information exchange and compact fusion.

The rest of this article is organized as follows. Section II details the network architecture of the proposed CCR-Net, including the CNN-based feature extractor and the CCR-based fusion module. Experiential results are conducted on two multimodal datasets, i.e., the HS-LiDAR Houston2013 data and the HS-SAR Berlin data in comparison with currently existing state-of-the-art multimodal classification methods in Section III. Finally, Section IV draws a conclusion with a possible future outlook.

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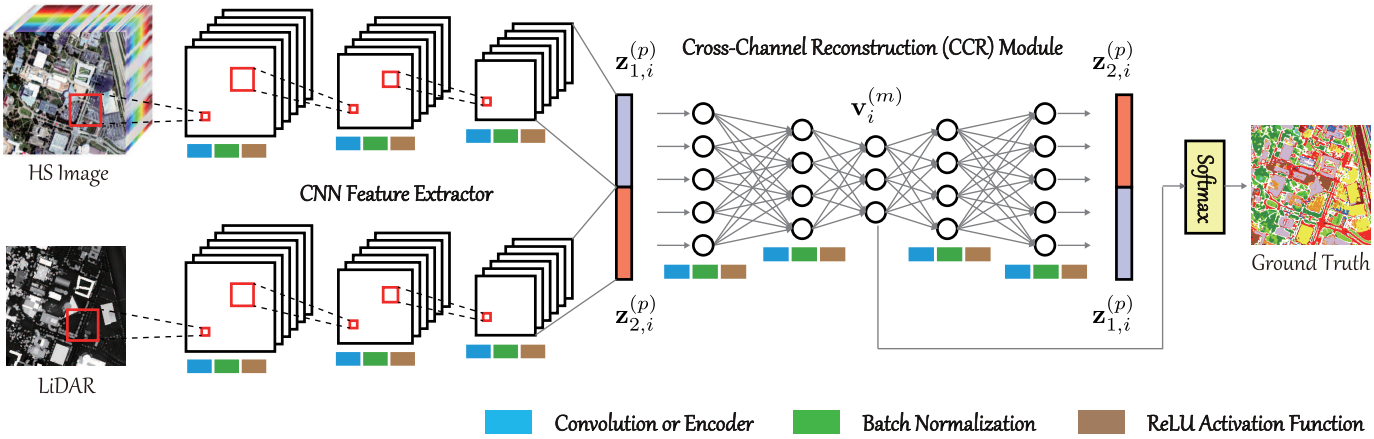



Fig. 2. Illustration to clarify the network architecture of the proposed CCR-Net for multimodal RS data classification.

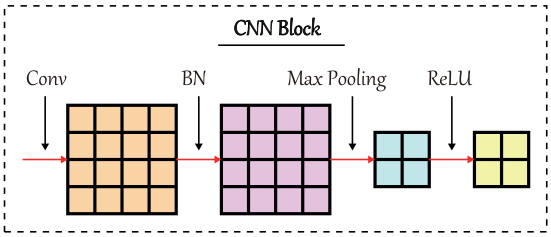



Fig. 3. Example of a CNN block in the proposed CCR-Net.

## II. Methodology

Fig. 2 illustrates the network architecture of the proposed CCR-Net. In detail, CCR-Net is nothing but a two-stream CNN, which starts with a feature extraction module for each modality and then feeds these features in a CCR module to fuse multimodal features in an encoder–decoder manner across different channels.

### A. Feature Extraction via CNNs

In recent years, CNN has been proven to be an effective tool to extract discriminative feature representations for RS data [44]. In this article, a CNN-based feature extractor is first used for each modality. Let $\mathbf{X}_s \in \mathbb{R}^{d_s \times N}, s = 1, 2$ with $d_1$ and $d_2$ channels by $N$ pixels be two different modalities, and the $l$th layer output in CNN is denoted as $\mathbf{Z}_s^{(l)}$. We then have

$$\mathbf{z}_{s,i}^{(l)} = \begin{cases} f\left(\mathbf{W}_s^{(l)}\mathbf{x}_{s,i} + \mathbf{b}_s^{(l)}\right), & l = 1 \\ f\left(\mathbf{W}_s^{(l)}\mathbf{z}_{s,i}^{(l-1)} + \mathbf{b}_s^{(l)}\right), & l = 2, \ldots, p \end{cases} \tag{1}$$

where $\mathbf{x}_{s,i}, s = 1, 2$ denotes the $i$th pixel in the $s$th modality and $p$ is the number of layers in CNN. $f$ is defined as the ReLU activation function with respect to the to-be-estimated network parameters: weights $\{\mathbf{W}_s^{(l)}\}_{l=1}^{p}$ and biases $\{\mathbf{b}_s^{(l)}\}_{l=1}^{p}$.

The extractor consists of two CNN blocks in each modality stream, and each block has a $3 \times 3$ convolutional layer, batch normalization (BN) layer [45], $2 \times 2$ max-pooling layer, and ReLU activation layer. Fig. 3 shows an example of a CNN block.

### B. Feature Fusion via CCR

Previous multimodal feature fusion approaches generally follow the concatenation-based fusion strategy [37], e.g., early fusion, middle fusion, and late fusion. The encoder–decoder architecture is a relatively new fusion fashion, which tends to yield more compact feature representations [37], [39]. Moreover, the CCR module follows beyond the encoder–decoder architecture by cross-channel prediction in order to make features extracted from different modalities fused in a more sufficient fashion. It is well known that self-supervised learning is capable of excavating the potential in feature representations without the need for large-scale labels. Inspired by the split-brain AEs [46] that follows a self-supervised learning strategy, the feature representations can be enhanced by using the cross-reconstruction-like manner. As a result, we attempt to design a feature-level cross-reconstruction module, i.e., CCR module, to blend the features of different modalities. More specifically, such a cross-reconstruction pattern in the CCR module can achieve effective information exchange and more compact fusion in the feature level. Fig. 4 illustrates the differences between four fusion strategies. The fusion process in the CCR module can be formulated as

$$\begin{aligned} \mathbf{v}_i^{(l)} &= g_{\text{en}}\left(\mathbf{z}_{s,i}^{(p)}, \mathbf{W}_v^{(p)}, \mathbf{b}_v^{(l)}\right) \\ &= f(\mathbf{W}_v^{(l)}\left[\mathbf{z}_{1,i}^{(p)}, \mathbf{z}_{2,i}^{(p)}\right] + \mathbf{b}_v^{(l)}), \quad l = p+1, \ldots, m \end{aligned} \tag{2}$$

where $\mathbf{v}_i^{(l)}$ denotes the pixelwise fused features in the $l$th layer. The function $g_{\text{en}}$ is defined as the encoder network in CCR, while the decoder part $g_{\text{de}}$ can be expressed by

$$\begin{aligned} \mathbf{u}_i^{(l)} &= g_{\text{de}}\left(\mathbf{v}_i^{(m)}, \mathbf{W}_u^{(l)}, \mathbf{b}_u^{(l)}\right) \\ &= f\left(\mathbf{W}_u^{(l)}\mathbf{v}_i^{(m)} + \mathbf{b}_u^{(l)}\right), \quad l = m+1, \ldots, n \end{aligned} \tag{3}$$

where $\mathbf{u}_i^{(l)}$ denotes the pixelwise reconstructed features in the $l$th layer. To perform the CCR, it is natural to learn the mapping from the input features $\{\mathbf{z}_{s,i}^{(p)}\}_{p=1}^{2}$ to the output reconstructed features $\mathbf{u}_i^{(n)}$ via networks. Fig. 5 details the components of the CCR module in our proposed CCR-Net.

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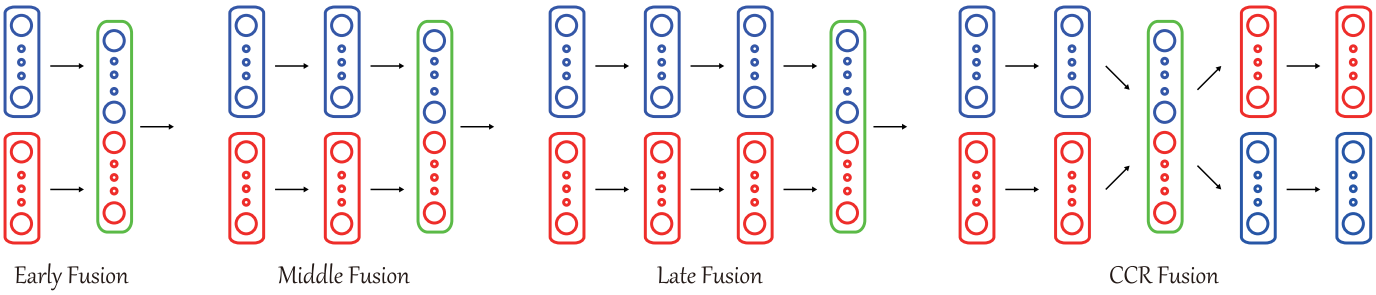



Fig. 4. Illustration for four different fusion strategies: early fusion, middle fusion, late fusion, and our CCR fusion.

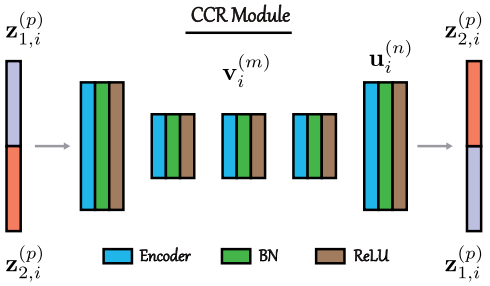



Fig. 5. Illustration of the CCR module in the proposed CCR-Net.

### C. Object Function

The network parameters in our CCR-Net can be updated by jointly optimizing the following overall loss function:

$$L = L_l + \alpha L_{\text{rec}} \tag{4}$$

where $\alpha$ is the penalty parameter to balance different terms in (4). In our case, the parameter $\alpha$ is experimentally and empirically determined to be 1, yielding a relatively stable performance. $L_l$ in (4) is the cross-entropy loss between the fused features $\mathbf{v}_i^{(m)}$ and one-hot labels $\mathbf{y}_i$, which can be written as

$$L_l = \frac{-1}{N}\sum_{i=1}^{N}\left[\mathbf{y}_i \log \mathbf{v}_i^{(m)} + (1-\mathbf{y}_i)\log\left(1-\mathbf{v}_i^{(m)}\right)\right] \tag{5}$$

and $L_{\text{rec}}$ is the CCR loss

$$L_{\text{rec}} = \sum_{i=1}^{N}\left|\left[\mathbf{z}_{2,i}^{(p)}, \mathbf{z}_{1,i}^{(p)}\right] - \mathbf{u}_i^{(n)}\right|_2^2 \tag{6}$$

where $\mathbf{u}_i^{(n)}$ denotes the output reconstructed features, as shown in (3) and Fig. 5. $L_{\text{rec}}$ aims to compute the loss regularized by $L_2$-norm between the reconstructed features $\mathbf{u}_i^{(n)}$ and the cross-channel features $[\mathbf{z}_{2,i}^{(p)}, \mathbf{z}_{1,i}^{(p)}]$.

## III. Experiments

### A. Data Description

*1) HS-LiDAR Houston2013 Data:* The datasets are provided for the IEEE GRSS DFC2013,[1] and the corresponding scene is acquired by the ITRES CASI-1500 imaging sensor over the campus of the University of Houston and its surrounding rural areas, Texas, USA. The HS image consists of 349 × 1905 pixels with 144 spectral channels ranging from 364 to 1046 nm at 10-nm spectral resolution, and the LiDAR data with only one band provide the height information for the same image region. There are 15 land cover land use categories in the studied scene. Table I details the scene information, including class name and the number of training and testing samples.

TABLE I
Description of the Investigated HS-LiDAR Houston2013 Datasets, Including the Object Name and the Corresponding Number of Training and Testing Samples

| No. | Class Name | Training Set | Testing Set |
|---|---|---|---|
| Class 1 | Healthy Grass | 198 | 1053 |
| Class 2 | Stressed Grass | 190 | 1064 |
| Class 3 | Synthetic Grass | 192 | 505 |
| Class 4 | Tree | 188 | 1056 |
| Class 5 | Soil | 186 | 1056 |
| Class 6 | Water | 182 | 143 |
| Class 7 | Residential | 196 | 1072 |
| Class 8 | Commercial | 191 | 1053 |
| Class 9 | Road | 193 | 1059 |
| Class 10 | Highway | 191 | 1036 |
| Class 11 | Railway | 181 | 1054 |
| Class 12 | Parking Lot1 | 192 | 1041 |
| Class 13 | Parking Lot2 | 184 | 285 |
| Class 14 | Tennis Court | 181 | 247 |
| Class 15 | Running Track | 187 | 473 |
| | Total | 2832 | 12197 |

*2) HS-SAR Berlin Data:* The datasets describe the Berlin urban and its rural neighboring area, which consists of an EnMAP HS image[2] simulated from the HyMap HS data and a corresponding Sentinel-1 SAR data covering the same region. The HS image contains 797 × 220 pixels and 244 spectral bands in the wavelength range of 400–2500 nm, while the SAR data are a dual-polarSAR, involving four bands (VV-VH). In addition, the ground-truth map is generated from the OpenStreetMap data, and the specific information regarding training and testing sets is listed in Table II. These datasets are openly and freely available from [32].

### B. Experimental Setup

*1) Evaluation Metrics:* In our experiments, the multimodal RS data classification performance is evaluated by computing three commonly used indices, i.e., *overall accuracy (OA)*, *average accuracy (AA)*, and *kappa coefficient* ($\kappa$). In general,

[1]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2]This image has a 30-m spatial resolution, which can be freely downloaded from the website: http://doi.org/10.5880/enmap.2016.002

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TABLE II
DESCRIPTION OF THE INVESTIGATED HS-SAR BERLIN DATASETS, INCLUDING THE OBJECT NAME AND THE CORRESPONDING NUMBER OF TRAINING AND TESTING SAMPLES

| No. | Class Name | Training Set | Testing Set |
|---|---|---|---|
| Class 1 | Forest | 443 | 54511 |
| Class 2 | Residential Area | 423 | 268219 |
| Class 3 | Industrial Area | 499 | 19067 |
| Class 4 | Low Plants | 376 | 58906 |
| Class 5 | Soil | 331 | 17095 |
| Class 6 | Allotment | 280 | 13025 |
| Class 7 | Commercial Area | 298 | 24526 |
| Class 8 | Water | 170 | 6502 |
| | Total | 2820 | 461851 |

the higher values for the three indices mean better classification performance in the RS image classification task. For those feature learning-based models, we first learn multimodal feature representations and then apply a classifier, e.g., nearest neighbor (NN) and random forest (RF), to make a quantitative performance comparison with other competitors. More specifically, detailed definitions for the three used indices, i.e., OA, AA, and $\kappa$, are given as follows:

$$\mathrm{OA} = \frac{N_c}{N_a} \tag{7}$$

$$\mathrm{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{N_c^i}{N_a^i} \tag{8}$$

and

$$\kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{9}$$

where $N_c$ and $N_a$ are the number of samples that are correctly recognized and the number of total to-be-classified samples, respectively. $N_c^i$ and $N_a^i$ denote the sample number for each class corresponding to $N_c$ and $N_a$, respectively. $P_e$ in $\kappa$ is defined as the hypothetical probability of chance agreement [47], which can be computed by

$$P_e = \frac{N_r^1 \times N_p^1 + \cdots N_r^i \times N_p^i + \cdots + N_r^C \times N_p^C}{N_a \times N_a} \tag{10}$$

where $N_r^i$ and $N_p^i$ are the numbers of real samples and predicted samples for each class, respectively.

*2) Comparison With State-of-the-Art Multimodal Algorithms:* To demonstrate the effectiveness of the proposed CCR-Net, several state-of-the-art multimodal algorithms are selected for both quantitative and qualitative comparison; they are single modalities (e.g., HS, LiDAR, and SAR), stacked features (directly using stacked multimodal data, e.g., HS-LiDAR and HS-SAR), CoSpace [27], LeMA [29], CapsNet [48], Co-CNN [38], and our CCR-Net.

*3) Implementation Details:* Our proposed CCR-Net was implemented on the Tensorflow programming platform. The Adam is adopted as the optimizer to update the network gradient with the batch size of 32. The size of the input patches in networks is $7 \times 7$ pixels. The initial learning rate is set to $1e-3$ with a 0.5 factor decaying after each three-tenth of total epochs. The network configuration of the CCR-Net consists of two parts: CNN-based feature extractor and CCR-based feature fusion. The former has two CNN blocks, including a convolutional layer, a BN layer, a max-pooling layer, and an ReLU layer (see Fig. 3). The latter is comprised of six fully connected blocks, where three blocks are the encoder module and the rest blocks are the decoder module. Please see more details in Fig. 5. Moreover, we empirically set the epochs to be 200 since we found that such iterations would basically enable the convergence of the model to a stable situation.

It is worth noting that we found by experiments that increasing the number of layers in a proper range (e.g., four or five layers in feature extraction, please see [37] for more details) would improve the classification performance to some extent but might bring more expensive computational cost. More importantly, the proposed CCR-Net in this article mainly aims to show the effectiveness and superiority of the CCR module in fusing multimodal feature representations of RS data rather than investigate the hyperparameter setting in networks. Therefore, we design the entire network architectures in compared networks and CCR-Net (apart from the particular modules) as consistent (or same) as possible for a fair comparison. As a tradeoff, we experimentally determined the aforementioned network configuration. In addition, regarding the amount of the parameter and complexity of the networks, it can be intuitively seen from the aforementioned introduction that the network architecture of the CCR-Net is basically identical to those compared networks, e.g., Co-CNN, for a fair comparison. This also means that either the quantity of parameters or network complexity is similar, except for the reconstruction decoder part (including three fully connected layers) in the CCR module.

### *C. Results and Analysis on the HS-LiDAR Houston2013 Data*

Table III quantitatively lists the classification results of different multimodal methods in terms of three commonly used indices: OA, AA, and $\kappa$, while Fig. 6 visually shows the corresponding classification maps obtained by these algorithms.

Overall, the classification results obtained with HS data are dramatically higher than those with LiDAR data using both NN and RF classifiers, while the O-Baseline methods by stacking multimodal features, i.e., O-Baseline-NN and O-Baseline-RF, obtain better classification performance compared to single HS and LiDAR data, indicating the effectiveness of the use of multimodal RS data. By aligning the multimodal training data, CoSpace obviously performs better than the O-Baseline with an increase of more than 3% OAs. Beyond the supervised method, LeMA fully considers labeled and unlabeled samples to find a better decision boundary, achieving higher classification accuracies. Unlike traditional subspace-based approaches, deep-learning-based models, e.g., CapsNet, Co-CNN, and our CCR-Net, are capable of extracting more discriminative feature representations and better capturing the potential relations between different modalities. This leads to better classification results with CapsNet and Co-CNN. Although the capsule unit is applied in CapsNet to extract richer features, the ability in feature fusion, particularly for heterogeneous RS data, remains limited. This, to some

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TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND $\kappa$ ON THE HS-LiDAR HOUSTON2013 DATASETS. THE BEST ONE IS SHOWN IN BOLD

| Methods | HS-NN | HS-RF | LiDAR-NN | LiDAR-RF | Stack-NN | Stack-RF | CoSpace | LeMA | CapsNet | Co-CNN | CCR-Net |
|---|---|---|---|---|---|---|---|---|---|---|---|
| OA (%) | 73.89 | 73.12 | 10.04 | 31.49 | 79.57 | 76.19 | 82.14 | 85.42 | 86.52 | 87.23 | **88.15** |
| AA (%) | 76.89 | 77.04 | 7.74 | 37.18 | 81.81 | 79.64 | 84.54 | 87.05 | 87.66 | 88.82 | **89.82** |
| $\kappa$ | 0.7192 | 0.7111 | 0.0320 | 0.2636 | 0.7793 | 0.7440 | 0.8062 | 0.8417 | 85.41 | 0.8619 | **0.8719** |
| C1 | 82.24 | 82.62 | **92.12** | 13.49 | 82.62 | 82.53 | 81.96 | 81.86 | 81.10 | 83.10 | 83.00 |
| C2 | 81.77 | 83.55 | 21.43 | 16.26 | 81.77 | 83.36 | 83.27 | 83.08 | 81.02 | **84.87** | **84.87** |
| C3 | 99.60 | 97.62 | 0.00 | 56.63 | **100.00** | 98.42 | **100.00** | **100.00** | 96.44 | 99.80 | **100.00** |
| C4 | 91.67 | 91.95 | 0.38 | 42.80 | 91.48 | 91.48 | 94.22 | **94.79** | 88.35 | 92.42 | 92.14 |
| C5 | 97.06 | 96.78 | 0.00 | 58.05 | 96.50 | 96.88 | 99.34 | 99.34 | **100.00** | 99.24 | 99.81 |
| C6 | 95.10 | **99.30** | 0.00 | 58.04 | 95.10 | 98.60 | **99.30** | **99.30** | 95.80 | 95.80 | 95.80 |
| C7 | 73.41 | 74.72 | 1.96 | 41.42 | 77.99 | 80.04 | 81.44 | 88.99 | 86.38 | 95.24 | **95.34** |
| C8 | 36.94 | 33.43 | 0.00 | 27.64 | 66.48 | 43.02 | 66.10 | 74.26 | **90.88** | 81.86 | 81.39 |
| C9 | 66.76 | 70.16 | 0.09 | 14.35 | 70.07 | 69.97 | 69.97 | 73.84 | 82.53 | **85.08** | 84.14 |
| C10 | 54.05 | 43.24 | 0.100 | 9.07 | 58.01 | 45.85 | 48.94 | 72.20 | **72.78** | 61.10 | 63.22 |
| C11 | 70.30 | 69.83 | 0.00 | 39.94 | 74.48 | 74.38 | 88.61 | 82.26 | 88.99 | 83.87 | **90.32** |
| C12 | 58.89 | 54.85 | 0.00 | 8.55 | 76.66 | 67.82 | 88.57 | 90.30 | 83.09 | 91.26 | **93.08** |
| C13 | 49.47 | 60.00 | 0.00 | 15.44 | 57.54 | 64.56 | 68.07 | 67.37 | 76.14 | **88.77** | 88.42 |
| C14 | 97.98 | 99.60 | 0.00 | 80.16 | **100.00** | **100.00** | **100.00** | **100.00** | 93.93 | 91.09 | 96.36 |
| C15 | 98.10 | 97.89 | 0.00 | 75.90 | 98.52 | 97.67 | 98.31 | 98.10 | 97.46 | 98.73 | **99.37** |

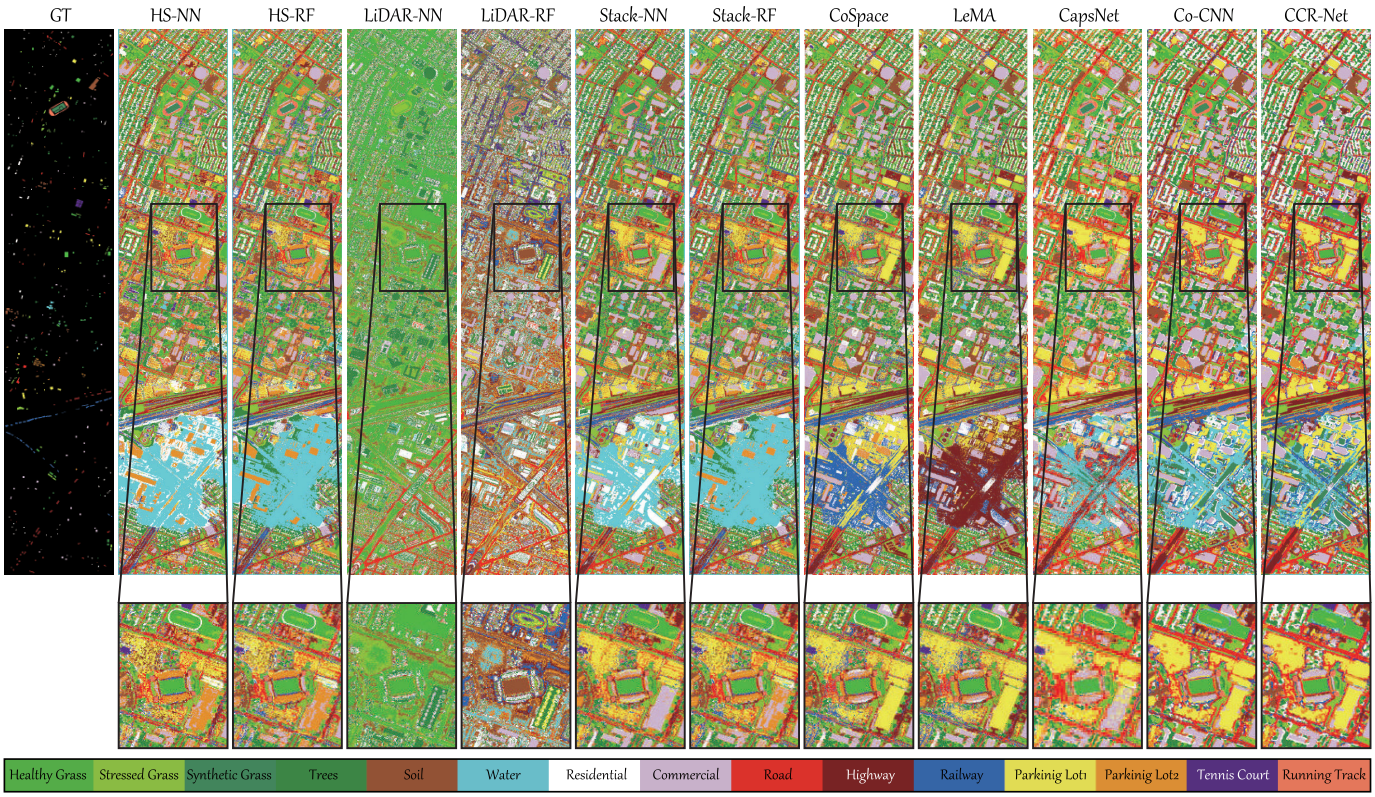



Fig. 6. Classification maps obtained by different multimodal algorithms on the HS-LiDAR Houston2013 datasets.

extent, results in slightly poor performance compared to Co-CNN. It is worth noting that the proposed CCR-Net outperforms other compared methods, which obtains the best performance in multimodal data fusion and classification. Moreover, the classification map generated by our CCR-Net can provide more realistic land cover land use labeling and better spatially aggregated results in comparison with other state-of-the-art methods. The materials on the bottom of this studied scene fail to be well recognized and identified due to the shadow region covered by clouds. However, remarkably, the proposed CCR-Net is, to some extent, capable of eliminating the effects of the cloud cover from the visual perspective. Furthermore, to better distinguish the differences of classification maps generated by different multimodal approaches, some close-ups are provided for better comparison, as shown in Fig. 6. Our CCR-Net tends to generate more realistic and smoother classification maps with less noisy pixels (wrong classification points).

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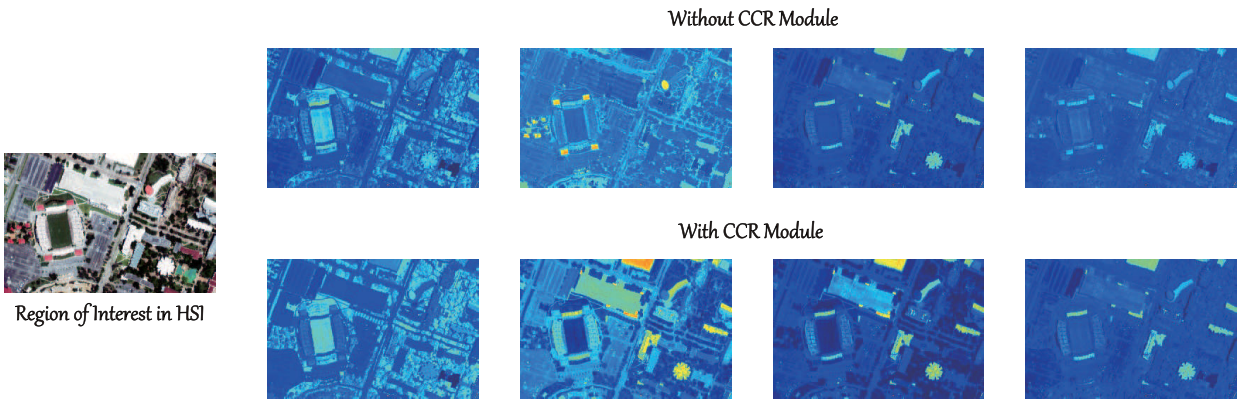



Fig. 7. Visual comparison to provide the contrast features without and with the CCR module in an ROI of HSI.

TABLE IV
ABLATION STUDY OF DIFFERENT FUSION STRATEGIES. THE BEST ONE IS SHOWN IN BOLD

| Fusion Strategy | Early | Middle | Late | CCR |
|---|---|---|---|---|
| OA (%) | 81.70 | 87.23 | 85.51 | **88.15** |
| AA (%) | 84.91 | 88.82 | 87.79 | **89.82** |
| $\kappa$ | 0.8013 | 0.8619 | 0.8427 | **0.8719** |

*1) Visual Comparison on Features:* Beyond the quantitative classification accuracy, we also verify the effectiveness of the CCR module in fusing multimodal features from the visual perspective. As shown in Fig. 7, the feature maps without and with the CCR module are visualized in a region of interest (ROI) of HSI, respectively. The features generated by the means of the CCR module are sharper in the edge and structure of objects than those without the CCR module, yielding more detailed and refiner appearances of fused features.

*2) Ablation Study on Fusion Strategy:* Early fusion, middle fusion, and late fusion are three kinds of main fusion strategies in multimodal deep fusion networks, where our CCR-Net is a variant of the middle fusion. To validate the effectiveness of the proposed CCR-based fusion module, a quantitative comparison is made among the early fusion, middle fusion (i.e., Co-CNN), late fusion, and CCR fusion (i.e., our CCR-Net) on the HS-LiDAR Houston2013 datasets. As listed in Table IV, the early fusion yields a relatively poor performance since the multimodal data coupling happens in the beginning, while the late fusion strategy focuses on the decision level and ignores the information exchange in the feature level to some extent. As a tradeoff, the middle fusion tends to achieve a better fusion effect. More importantly, the CCR-Net based on the newly designed CCR fusion performs better than the aforementioned baseline fusion strategies, further improving the classification performance on the basis of the middle fusion (around 1% OA).

*3) On the Varying Training Size:* In this section, we investigate the stability and sensitivity of the proposed CCR-Net with the varying number of training samples on the Houston2013 datasets, as shown in Fig. 8. It is relatively obvious that, despite only using a 10% total of training samples, our CCR-Net still brings around 83% OA. With the gradually increasing percentage of the used training samples, the classification accuracies are able to promptly reach the optimal value and remain basically stable. In particular, the change of the OA values is very small, nearly negligible when the percentage is larger than 50%. This, to a great extent, demonstrates the stability of the CCR-Net with regard to the different percentages of training samples.

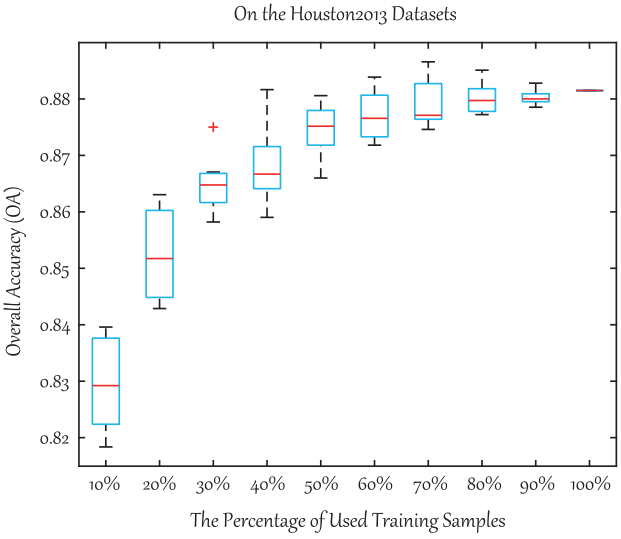



Fig. 8. Classification performance of the proposed CCR-Net with the varying number of training samples on the Houston2013 datasets.

### *D. Results and Analysis on the HS-SAR Berlin Data*

Similarly, the quantitative classification performance on the HS-SAR Berlin datasets is listed in Table V, and the corresponding classification maps are visualized in Fig. 9.

There is a basically identical trend in classification performance to that on the HS-LiDAR Houston2013 datasets. That is, the joint use of multiple RS data sources can effectively improve the classification accuracies, and single HS data are more discriminative compared to the SAR data, which can be better used to identify the materials. The features learned via joint feature learning models, e.g., CoSpace and

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TABLE V

QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA, AA, AND $\kappa$ ON THE HS-SAR BERLIN DATASETS. THE BEST ONE IS SHOWN IN BOLD

| Methods | HS-NN | HS-RF | SAR-NN | SAR-RF | Stack-NN | Stack-RF | CoSpace | LeMA | CapsNet | Co-CNN | CCR-Net |
|---|---|---|---|---|---|---|---|---|---|---|---|
| OA (%) | 50.30 | 54.94 | 21.98 | 32.86 | 51.71 | 55.82 | 64.53 | 66.71 | 66.55 | 68.51 | **69.85** |
| AA (%) | 58.39 | 58.47 | 21.18 | 34.65 | 59.04 | 58.50 | 62.51 | 62.35 | 61.18 | 62.34 | **66.99** |
| $\kappa$ | 0.3709 | 0.4134 | 0.0759 | 0.1790 | 0.3838 | 0.4208 | 0.5093 | 0.5312 | 0.5277 | 0.5476 | **0.5716** |
| C1 | 71.45 | 79.29 | 24.47 | 32.53 | 73.57 | 79.31 | 85.09 | 85.11 | 84.96 | 84.09 | **85.93** |
| C2 | 41.19 | 48.68 | 19.86 | 30.72 | 42.93 | 50.46 | 61.60 | 64.84 | 65.22 | **68.48** | 68.07 |
| C3 | 40.37 | 40.84 | 25.53 | 36.60 | 38.72 | 41.64 | 51.18 | 48.94 | 48.42 | 49.09 | **53.17** |
| C4 | 68.07 | 66.68 | 31.79 | 44.07 | 69.23 | 65.82 | 75.44 | 80.04 | 80.80 | 79.43 | **82.62** |
| C5 | 81.05 | 67.15 | 31.61 | 52.00 | 81.01 | 66.94 | 82.50 | 80.66 | 69.18 | 81.25 | **85.10** |
| C6 | **63.89** | 61.19 | 13.32 | 33.37 | 62.89 | 61.01 | 54.66 | 54.07 | 55.08 | 50.68 | 63.02 |
| C7 | 35.31 | **37.01** | 14.96 | 13.04 | 36.83 | 35.84 | 28.81 | 27.40 | 26.12 | 26.16 | 29.23 |
| C8 | 65.76 | 66.90 | 7.91 | 34.88 | 67.13 | 66.98 | 60.78 | 57.75 | 59.69 | 59.52 | **68.78** |

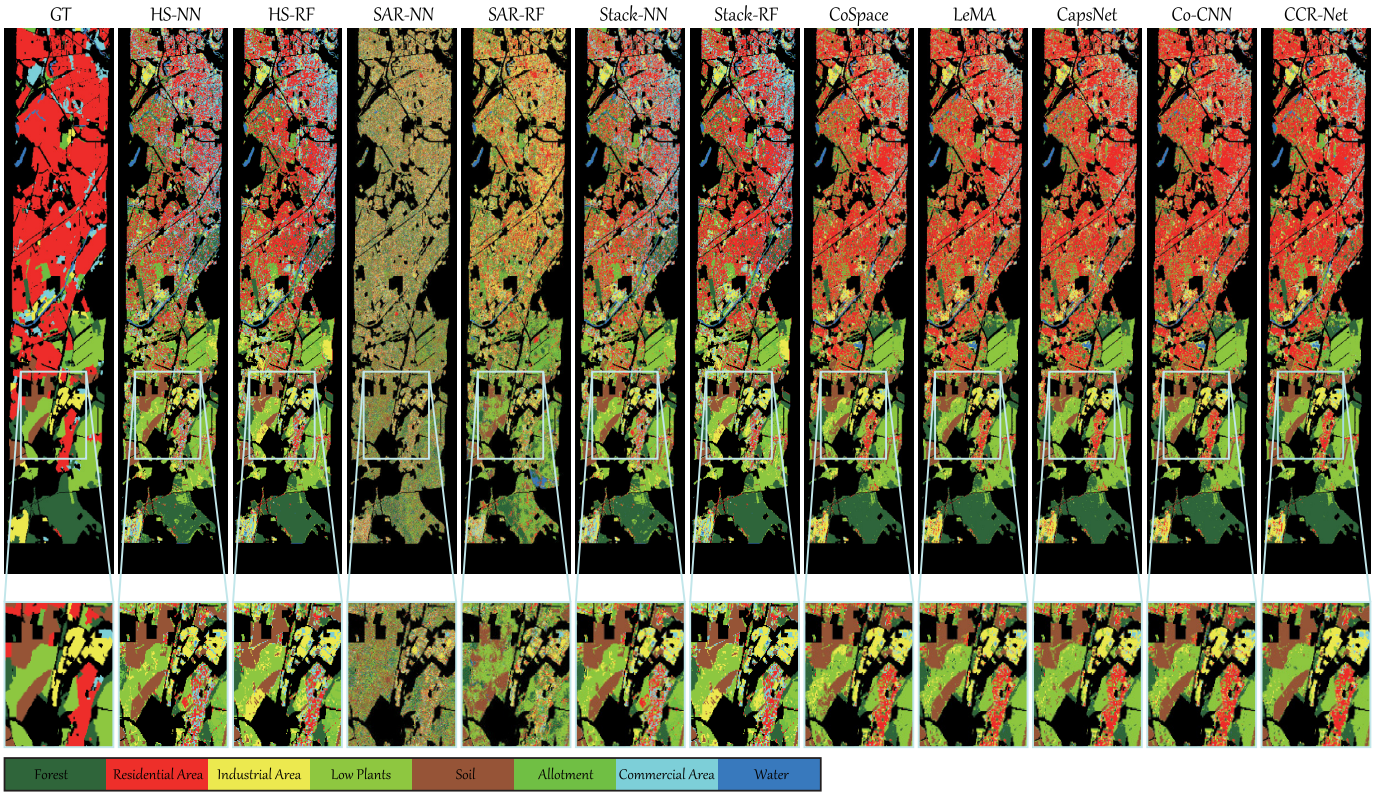



Fig. 9. Classification maps obtained by different multimodal algorithms on the HS-SAR Berlin datasets.

LeMA, more tend to find a better decision boundary of classifiers. Deep-learning-based methods have a more powerful ability in representing different modalities, thereby tending to complete the classification task. Visually, the classification maps generated by deep learning, particularly our proposed CCR-Net, are smoother than other competitors due to the use of spatial information and more effective transferring of other modalities. Similarly, the CCR-Net tends to yield better visual verisimilitude, particularly for the *Residential* and *Forest* classes that can be identified more effectively and accurately and provide better visual cues. As shown in Fig. 9, the locally detailed classification maps are given by different compared algorithms, where we can see that our CCR-Net obtains a more accurate classification result that is closer to the GT's, particularly for those classes, e.g., *soil* and *low plants*.

## IV. CONCLUSION

In this article, we propose a novel multimodal RS image classification network, called CCR-Net, by introducing a CCR module. The use of CCR makes the features of different modalities obtained by the CNN extractor fused in a more compact way, tending to lead to a better classification performance compared to traditional concatenation- and alignment-based fusion methods. This further demonstrates the effectiveness and superiority of the proposed CCR-Net in the classification task. Although the newly proposed CCR fusion

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module in networks provides new insight into the multimodal land cover classification task, its practicability and application ability need to be further developed, particularly for complex real scenes and more modality data (e.g., more than three modalities).

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